



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 2 Report

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Experiment 10: Computer Vision – CNNs & OpenCV during the M-Tech Production & Marketing AI/ML Internship Program 2026. The experiment introduced the field of Computer Vision and explained how Convolutional Neural Networks (CNNs) and OpenCV are used for image processing and recognition tasks. It provided practical knowledge of how computers analyze and understand visual information.

2. Objectives

- Understand the fundamentals of Computer Vision.
 - Learn the architecture and working of Convolutional Neural Networks (CNNs).
 - Explore image processing techniques using OpenCV.
 - Understand feature extraction through convolution and pooling layers.
 - Study real-world applications of Computer Vision.
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3. Task Description

The experiment focused on understanding Computer Vision techniques and the working of CNN models for image classification. It also introduced OpenCV as a powerful image processing library used for loading, displaying, and manipulating images. The experiment demonstrated how convolution and pooling layers extract meaningful image features for accurate predictions.

4. Work Done

During this experiment, I learned the fundamentals of Computer Vision and how machines analyze images using deep learning models. I studied the architecture of Convolutional Neural Networks, including convolution layers, pooling layers, fully connected layers, and output layers. I explored the use of OpenCV for basic image processing tasks such as reading images, resizing, converting images to grayscale, and detecting image features. I also learned how CNNs automatically extract important visual patterns from images and use them for classification. Practical examples helped me understand how Computer Vision is applied in facial recognition, medical imaging, autonomous vehicles, and object detection systems.

5. Challenges Faced

- Understanding convolution and pooling operations.
 - Learning how CNN layers extract image features.
 - Becoming familiar with OpenCV image processing functions.
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6. Key Learnings

- Computer Vision enables machines to interpret visual information.
- CNNs automatically learn image features through convolution layers.
- OpenCV provides powerful tools for image processing.
- Pooling layers reduce image dimensions while preserving important features.
- Computer Vision is widely used in healthcare, security, robotics, and self-driving cars.

7. Conclusion

This experiment enhanced my understanding of Computer Vision and Convolutional Neural Networks. I learned how CNNs process images and how OpenCV supports image manipulation and preprocessing. The experiment strengthened my practical knowledge of modern AI applications in image analysis and recognition

Intern Signature:



Supervisor:

Name: Ghazi Muhammad Abdullah

Name:

Date: 14 July 2026

Designation: